SET: III

AMERICAN INTERNATIONAL UNIVERSITY – BANGLADESH

Faculty of Science and Technology

Department of Mathematics

MAT 3103: Computational Statistics and Probability (All Sections)

Final Examination Fall: 2019-20

Total Marks: 40 Time: 2 hours

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Instructions	⇨ Question **1** is **compulsory** (must be answered).
	⇨ Answer any **four** from the remaining questions (Question **2** to Question **6)**

1. Answer **any four** from the followings: **(4 × 2 = 8)**

a) Given, $M_x(t) = exp\ (\mu t + \frac{1}{2}\sigma^2 t^2)$. Calculate mean and variance of X.

b) Explain simple random sampling with a real-life example.

c) Giving an example, show when the regression coefficient is four ($b = 4$).

d) Under what conditions a counting process becomes Poisson process?

e) A company claims that its batteries have a life of 100 hours. You try to verify this for a sample of size 121 with mean 97 hours and standard deviation 3 hours.

2. a) Time taken to assemble a car in a plant follows $N(15, 50)$. What is the probability that a car can be assembled at this plant in a period of time (**a**) less than 20 hours? and (**b**) between 12 to 22 hours? **4**

b) The average time needed to send a signal from a server is 5 minutes. Find the probability that a signal will be sent (a) before 10 minutes, and (b) after 10 minutes. **4**

3. a) Customers enter to a mobile repairing shop from 9 – 10 and from 10 – 11 with probability 0.8. Customers also enter to the shop from 10 – 11 but not from 9 – 10 with probability 0.5. Find the probability that the customers will enter to the shop up to 2 PM starting from 9 am. **4**

b) Customer enters a bank at Poisson rate λ = 25 per hour. Find the probability that the elapsed time between the entrance of 7th and 8th customer is (i) more than 1 hour, (ii) less than 1 hour. **4**

4. a) Determine a sample of size n to estimate a population mean with margin of error 0.5 at 95% level of confidence, where variance of the population observations is 6.35. **2**

b) Number of emails received by a server in various days are: 5, 8, 7, 1, 7, 6, 9, 1, 4, 2, 7, 7.

(i) Select a sample of size $n = 3$ by simple random sampling. **2**

(ii) Estimate mean number of emails along with standard error. **4**

5. a) Among 157 African-American men, the mean systolic blood pressure was 146 mm Hg with a standard deviation of 27. We wish to know if on the basis of these data, we may conclude that the mean systolic blood pressure for a population of African-American is 140. **2**

b) Out of 25 students, 18 are male. Is the overall proportion of male students 0.7 in AIUB? **2**

c) Do you think that the employment facility for both the universities is similar? **4**

University	Number of graduates employed in computer related job
A	x_1 : 18, 16, 15, 20, 18, 15, 12
B	x_2 : 20, 14, 12, 22, 16, 14, 15, 10, 12, 18, 10

6. The following are the data representing age and BP of some selected persons.

Age $(x, in\ year)$	25	30	35	30	32	40	45	40	36	35
BP $(y, in\ mm\ Hg)$	75	80	85	90	95	85	100	90	85	80

a) Compute correlation coefficient. **2**

b) Do you think that BP increases significantly with the increase in age? **2**

c) Fit a regression line of y on x. **2**

d) Test the significance of regression. **2**

List of formulas		
$v(\bar{x}) = \frac{N-n}{Nn} s^2$	$v(p) = \frac{N-n}{N(n-1)} pq$	$s^2 = \frac{1}{n-1}\left[\sum x^2 - \frac{(\sum x)^2}{n}\right]$
$v(\hat{X}) = N^2 v(\bar{x})$	$s.e.(p) = \sqrt{v(p)}$	$\bar{x} \pm t_{n-1} * s.e.(\bar{x})$
$s.e.(\bar{x}) = \sqrt{v(\bar{x})}$	$\hat{X} = N\bar{x}$	$\hat{X} \pm t_{n-1} * s.e.(\hat{X})$
$n = \frac{z^2\, v(X)}{d^2}$	$n = \frac{z^2\, pq}{d^2}$	$s.e.(\hat{X}) = \sqrt{v(\hat{X})}$
$P(X > x) = e^{-\frac{x}{\lambda}}$	$P(X < x) = 1 - e^{-\frac{x}{\lambda}}$	$P(x_1 < X < x_2) = e^{-\frac{x_1}{\lambda}} - e^{-\frac{x_2}{\lambda}}$
$P(T > t) = e^{-\lambda t}$	$P(T < t) = 1 - e^{-\lambda t}$	$P(t_1 < T < t_2) = e^{-\lambda t_1} - e^{-\lambda t_2}$
$P(X > x) = e^{-\frac{x^2}{2\sigma^2}}$	$P(X < x) = 1 - e^{-\frac{x^2}{2\sigma^2}}$	$P(x_1 < X < x_2) = e^{-\frac{x_1^2}{2\sigma^2}} - e^{-\frac{x_2^2}{2\sigma^2}}$
$z = \frac{\bar{x} - \mu_0}{\sqrt{\frac{s^2}{n}}}$	$t = \frac{\bar{x} - \mu_0}{\sqrt{\frac{s^2}{n}}} \sim t_{n-1}$	$\chi^2 = \sum\sum \frac{O_{ij}^2}{E_{ij}} - n \sim \chi^2_{(r-1)(c-1)}$
$z = \frac{p_1 - p_2}{\sqrt{PQ\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$	$z = \frac{p - p_0}{\sqrt{\frac{p_0 Q_0}{n}}}$	$t = \frac{\overline{x_1} - \overline{x_2}}{\sqrt{s^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim t_{(n_1-1)+(n_2-1)}$
$P = \frac{a_1 + a_2}{n_1 + n_2}$	$r = \frac{SP(xy)}{\sqrt{SS(x)\, SS(y)}}$	$s^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{(n_1-1) + (n_2-1)}$
$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{n-2}$	$SS(y) = \sum y^2 - \frac{(\sum y)^2}{n}$	$\chi^2 = \sum \frac{O_i^2}{E_i} - n \sim \chi^2_{(k-1)},\ E_i = \frac{n}{k}$
$SS(x) = \sum x^2 - \frac{(\sum x)^2}{n}$	$SP(xy) = \sum xy - \frac{\sum x \sum y}{n}$	$\chi^2 = \frac{n(ad-bc)^2}{(a+b)(a+c)(b+d)(c+d)} \sim \chi_1^2$
$\hat{y} = a + bx$	$TSS = \sum\sum {y_{ij}}^2 - CT$	$SS\,(Error) = TSS - SS\,(Factor)$
$z = \frac{\overline{x_1} - \overline{x_2}}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$	$t = \frac{b}{\sqrt{\frac{s^2}{ss\,(x)}}}$	$SS\,(Factor) = \sum \frac{{y_i}^2}{n_i} - CT$
$b = \frac{SP(xy)}{SS(x)}$	$MS = \frac{SS}{df}, CT = \frac{G^2}{n}$	$s^2 = \frac{ss(y) - b\, sp(xy)}{n-2}$

Random number		**t Table**				**χ^2 Table**		**Normal Table**	
12212	01885	**df**	**value**	**df**	**value**	**df**	**value**	**z**	**$P(z)$**
12294	10179	1	12.706	11	2.201	1	3.841	0.14	0.5557
11584	17594	2	4.303	12	2.179	2	5.991	0.71	0.7611
16986	15206	3	3.182	13	2.160	3	7.815	0.99	0.8389
16141	09232	4	2.776	14	2.145	4	9.488	- 0.42	0.3372
09294	41214	5	2.571	15	2.131	5	11.071	0.17	0.5675
12584	54259	6	2.447	16	2.120			- 0.83	0.8033
16986	20888	7	2.365	17	2.110	**F Table**		0.42	0.6628
19278	16278	8	2.306	18	2.101	$F_{4,20}$	2.87	- 0.33	0.3707
10553	01177	9	2.262	19	2.093	$F_{3,16}$	3.24	- 0.25	0.4013
09876	16141	10	2.228	20	2.086	$z_{0.05} = 1.96$			